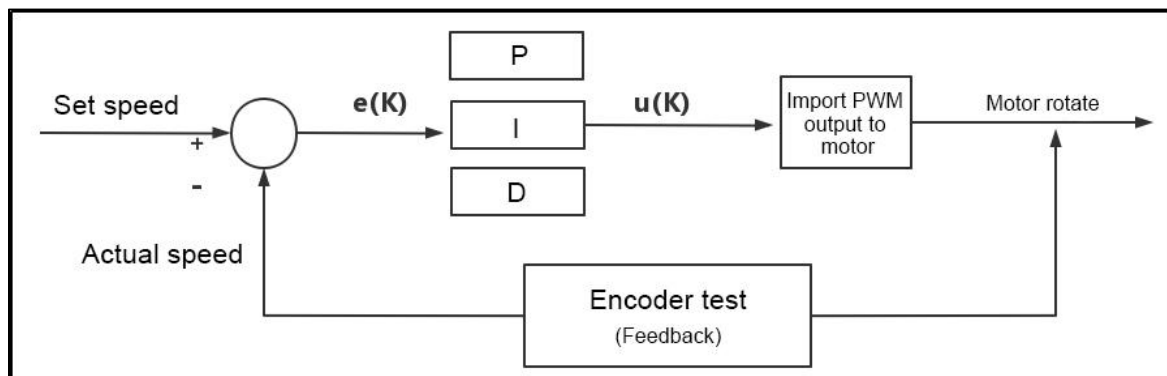


Speed PID Control Basics

The PID speed control can make motor reach a specific speed accurately and quickly. The speed information of motor is measured based on the number of pulses acquired per unit time (M-method velocimetry is used) and compared with the target value to get the control deviation, and then the deviation is controlled by proportional (P), integral (I), and differential (D), aiming to minimize the deviation and bring it closer to zero.

The construction diagram of motor speed PID control is shown below:



You can refer to the following formulate:

$$PWM += K_p[e(k) - e(k-1)] + K_i \cdot e(k) + K_d[e(k) - 2e(k-1) + e(k-2)]$$

$e(k)$: current deviation

$e(k-1)$: previous deviation

$e(k-2)$: previous deviation before the last deviation

PWM represents incremental output

Only PI control is applied in closed-loop control system for speed control, thus PID controller can be simplified to the following equation:

$$PWM += K_p[e(k) - e(k-1)] + K_i \cdot e(k)$$